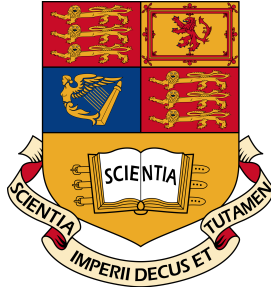


Department of Aeronautics
Imperial College London



Earth - Titan Rocket Propulsion System Design Report

Module: Year 3 Group Design Project
Project: Titan Atmospheric Flier
FATT: Propulsion and Powerplant
Author: Alex Gillespie
CID: 00747114
Date: 15/06/2015

This report forms part of a series of works which propose a preliminary design of an aerial reconnaissance vehicle which will perform scientific measurements over the surface of Titan, Saturn's largest moon, for at least 12 months. In particular this report outlines the design of a system that will allow the aerial vehicle and an orbiter vehicle to be delivered from Earth to Titan. The proposed system consists a two-staged launch vehicle (The Falcon 9 v1.1), a solar - ion propulsion upper stage (integral to the orbiter vehicle), a monopropellant secondary propulsion system and monopropellant reaction control system.

"A thousand things can happen when you light a rocket engine. And only one of them is good."
- Anonymous

Chapter 1

Introduction

1.1 Context and Motivation

Titan is the largest moon of Saturn and the second largest moon in the solar system - larger even than the planet Mercury. It's main distinction is that it is the only natural satellite with an atmosphere - a nitrogen-rich atmosphere with a density 5 times greater than ours. It is also, notably, the only other body to have surface liquid - lakes of methane and ethane in the northern polar regions. These features lead the moon to be of great scientific interest - indeed many scientists theorize that information from the atmosphere and surface could provide answers to the natural history of our own planet. Titan has had close approaches from several deep space probes but has only been analysed in any detail by the Cassini-Huygens mission - the Huygens probe descended through its atmosphere to reach the surface.

Means of analysing celestial atmospheres and surfaces (such as those of Venus and Mars) can typically be performed with instruments aboard satellites. Titan, however is shrouded in a thick layer of organic haze which can only be penetrated by a limited range of signals - meaning there is still a great deal of mystery regarding the details of the atmosphere and surface. This forms the main motivation behind this project - an unmanned aircraft to operate in the atmosphere and record measurements that an orbiting probe cannot. The proposed aircraft is 207kg, propeller-driven and nuclear-powered and is designed to operate for at least 12 months where it will survey the landscape and determine the composition and properties of the atmosphere. A second proposed vehicle as part of this system is an orbiter which will orbit the moon, relaying communications between the aircraft and Earth.

This report outlines the design processes and choices for a propulsion system which will allow this 2-vehicle system to be transported from the surface of the Earth to an orbit around Titan and also for the orbiter to correct and maintain its orbit while in-situ.

1.2 Rocket Propulsion

Rocket propulsion is a type of jet propulsion which provides thrust to a flying object by imparting momentum to stored matter - known as propellant. Commonly, this momentum is supplied by releasing chemical energy within the propellant(s), however other means are possible - such as with heat or from an electric charge. The main classification of Chemical rocket engines is the type of stored propellant - solid, liquid or gas - which each have distinct advantages which are summarised in table 1.1.

These engine types comprise the vast majority of all rocket engines and were the three main types considered for the mission. The use of liquid and solid fuel is used primarily in launch vehicles due

Table 1.1: Comparison of propellant types [1]

Propellant	Advantages	Disadvantages	Applications
Solid	- Relatively simple design - Very high thrust	- Cannot be throttled - Cannot be shut down - Lower efficiency	- Solid rocket boosters (SRBs) - Third stage main engines
Liquid	- Very high thrust - Fully throttleable - Light-weight tanks	- Complex, expensive design - Engines are heavy	- First stage main engines - Second stage main engines
Gas	- Simple, lightweight design - Very reliable - Capable of very low thrust	- Thrust is limited - Tanks are heavy	- Upper stage main engines - Reaction control thrusters

to the enormous thrust required to act against the atmosphere and the gravitational field. Liquid rocket engines use two propellants - an oxidiser (usually liquid oxygen) and a fuel (for example liquid hydrogen or RP1). The two propellants react together in the combustion chamber to produce extremely hot, fast-moving gas which channeled out the nozzle to produce thrust. Liquid fuel engines utilise turbopumps - complex turbomachines which use the energy from the combustion to vaporise the propellants before they are fed into the combustion chamber. Turbopumps are heavy, expensive and usually only have an operating lifetime of less than 10 minutes so they are only really used where there is a significant requirement [1].

Solid fuel rockets use a solid matrix of fuel and oxidiser which react together with an ignition charge. The matrix is packed into a chamber with a hollow cross-section. The shape of this cross section is specially designed to match the required burn rate distribution over time. Once the engine is ignited, it cannot be stopped or throttled. This limitation means that solid fuel engines cannot be the primary source of propulsion on a launch vehicle. Instead they are used as boosters which provide additional lift-off thrust to the main (turbopump) engines. The use of solid rocket boosters (SRBs) can provide flexibility in the lifting capability of the launch vehicle (for example, a manufacturer might specify that 6, 4, 2 or 0 SRBs can be used) [1].

Gas rocket engines are discussed in section 3.1.

1.3 Propulsion Analysis and Sizing

The main architecture of a typical propulsion system consists of a launch vehicle (to reach an orbit around the earth) and an upper stage (to reach the destination). Also on board the upper stage is often a reaction control system (see section 4.1). The launch vehicle for this mission can be considered 'off the shelf' i.e. a launch system is very rarely designed with one particular mission in mind (a notable exception being the Saturn V) and is instead chosen based on mission specifications. The upper stage and RCS however are very dependent on the mission specifications and must be designed with it in mind.

The main purpose of the analysis is to answer the question: What is the total mass and the mass of propellant need to take this spacecraft, with these engines to this destination? This outlines 3 main inputs.

Firstly, the matter of 'this spacecraft' basically means the dry mass - that is the mass of everything without the fuel. This contains an extensive list of systems such as: the solar panels, the engines themselves, the communications and the flier itself. Mass values of these inputs were largely answered by my colleagues in this project [2] [3] [4].

Secondly, the most important thing when considering the engine is its *specific impulse*. This can be thought of as the efficiency of the engine: it is a measure of how much force can be supplied per unit mass of propellant:

$$I_{sp} = \frac{F}{g_0 \dot{m}} \quad (1.1)$$

This value varies significantly for different engines and was the main design driver of the upper stage propulsion (chapter 3). Other than specific impulse, parameters such as the mass of the engines and the type of fuel also have an influence in design.

Lastly, the destination. The measure of how 'far' one can go in orbital mechanics is not a distance but instead a property known as ΔV . Literally meaning 'change in velocity' it is the measure of the impulse needed during a maneuver to get from one orbit to another. One can traverse the solar system using a series of these maneuvers to change from one orbit to the next. How much ΔV you chose to have available depends on the sum of ΔV needed to perform these maneuvers. An example of ΔV needed for particular maneuvers are shown below.

Table 1.2: ΔV for certain maneuvers

Maneuver	ΔV [m/s]
Earth to LEO	9400
LEO to GEO	3910
LEO to HCO	3210
LEO to Saturn insertion	7710

The value quoted for the LEO to Saturn insertion maneuver involves a direct *Hohmann transfer* which is the most energy efficient way to move from one circular orbit to the next. That is, however if there are no other planetary bodies. Instead, planets can be used for gravity assists which provide ΔV without spending fuel. Our mission uses a gravity assist from the Earth. The total estimated ΔV for the mission from LEO is 6630m/s [4].

Combining these inputs to find a total mass value was performed in two ways. First, the ideal rocket equation was used:

$$\Delta V = g_0 I_{sp} \ln \frac{m_t}{m_d} \quad (1.2)$$

This simple equation relates the ΔV and the specific impulse to the fraction of total mass (m_t) to dry mass (m_d). Even though the equation makes a lot of assumptions, it is actually very accurate in predicting deep space maneuvers [1].

Secondly, the software *GMAT* (General mission analysis tool) was used. It is a powerful program developed by NASA to analyze spacecraft and orbital dynamics [4].

The sizing process is discussed in more detail in chapter 5.1.

Chapter 2

Launch Vehicle

2.1 Selection Criteria of the Launch Vehicle

The function of a launch vehicle is to take payloads from Earth into space. There is a variety 'stock' launch vehicles in use today - mostly differing by the type of propellants and their lift capability. Since the vast majority of launches are used to put satellites into orbit, the lift capability is usually given by the maximum mass of payload that can be taken to Low Earth Orbit (LEO) and to Geostationary Transfer Orbit (GTO). For missions seeking to send a spacecraft beyond an Earth-centric orbit (such as ours), the maximum payload that can be sent into a Heliocentric Orbit (HCO) is often also quoted.

A launch vehicle can be selected based on the mass and destination of the payload. Generally, if one is looking to save on costs, it is clear that the launch vehicle should be appropriately sized as larger rockets (with greater lift capability) are, generally, more expensive to build and fuel. Another important consideration is the reliability of the launch vehicle based on the proportion of launches that are successful. As one might expect launch failures are expensive for everyone involved - for this reason there is a rigorous testing process that all launch vehicles and their components must go through; meaning that all launch vehicles in use today have very strong track records - close to 100%. A further consideration should be made regarding the internal dimensions of the payload fairing - which the spacecraft (orbiter and entry capsule) will be stowed inside before the launch. These fairings are not bespoke designs - rather the manufacturer specifies a limited range of options (typically less than 5) which must be able to contain the payload. Below is a comparison of various launch vehicles considered for the mission:

Table 2.1: Launch Vehicle Comparison

	Delta IV[5]	Titan IV[6]	Atlas V/Centaur[7]	Falcon 9v1.1[8][9]
Propellant type	LH2/LOX	-N204/A50 -SRB	-RP1/LOX (S1) -LH2/LOX (S2)	RP1/LOX
Payload to LEO [kg]	9,420-28,790	21,680	9,800-18,810	13,150
Payload to GTO [kg]	4,440-14,220	6,350	4,750-8,900	4,850
Payload to HCO [kg]	Unspecified	5,660	Unspecified	2,473
Launch cost [\$M]	350	432	100	61.7
Success rate	28/29	35/39	53/54	13/13

2.2 Falcon 9 v1.1

For the mission it was considered that the Falcon 9v1.1 would be the most suitable launch vehicle. The main driver was the price - the launch costs are vastly cheaper than American and European competitors and indeed cheaper even than Russian and Chinese competition. The economical savings of the Falcon 9 is attributed to the simplicity of the design. For example, only one type of fuel and oxidiser are used and the engine for the first and second stage are very similar in design (Merlin 1D and Vacuum). SpaceX, the manufacturer of the Falcon 9, is also pursuing reusable operation of the launch vehicle - with which they have had some recent success. This highlights the possibility that the company could be operating reusable hardware which would lead to even greater cost savings [10].

The main cause of launch failure is premature engine shutdown - normally in the first stage. The falcon 9 is named as such because it has 9 main engines in its first stage. This offers a degree of redundancy which would allow it to take off with only 8 engines operational and would still be able to complete its mission with 7 engines after 90 seconds. This is a unique feature and shows strong potential reliability of the launch vehicle.

Important specifications are shown in table 2.2 and the geometry of the fairing can be found in the appendix.

Table 2.2: Falcon 9 v1.1 specifications

Maximum take-off mass [kg]	505,846 + payload
Total height [m]	68.4
Launch sites	Cape Canaveral, FL Vandenberg AFB, CA
Fairing height (int) [m]	11.4
Maximum fairing diameter (int) [m]	4.6
Engines - first stage	9 Merlin 1D
Engines - second stage	Merlin Vacuum

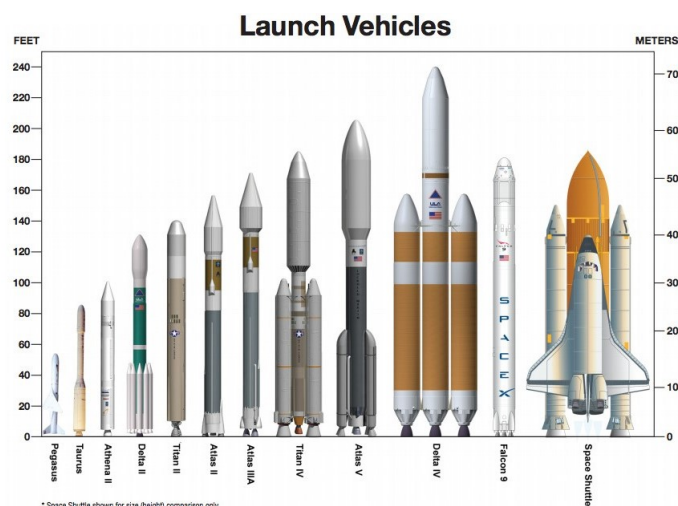


Figure 2.1: A comparison of launch vehicles (Falcon 9 picture with Dragon capsule rather than payload fairing). A larger version can be found in the appendix.

Chapter 3

Upper Stage Propulsion and Solar Power

3.1 Upper Stage Propulsion and Pressure-Fed systems

The role of an upper stage is to provide propulsion beyond that of the launch vehicle. With regards to deep space missions this upper stage is often fully integrated into the payload and provides various propulsion tasks throughout the mission life from interplanetary travel to orbital maintenance. The propulsion systems used in deep space are vastly different from those used in the launch vehicle - both in terms of size and style. While a launch vehicle is concerned with producing vast amounts of force to fight against the work of the gravitational field and atmosphere, in deep space the propulsion system is designed to be light-weight and efficient.

Historically, deep space propulsion has most commonly been produced by pressure-fed engines which uses a pressurant gas (usually helium) to drive the propellant(s) into the combustion chamber. The lower thrust requirements mean lower flow rates through the engine and therefore heavy, complex and less reliable turbopumps don't have to be used. A simplified schematic of a pressure-fed system can be found in the appendix

Pressure-fed systems can be split into two main categories: monopropellant and bipropellant. The former uses a single propellant that rapidly decomposes in the presence of a catalyst while the latter reacts two propellants together like a conventional liquid fuel rocket engine. Both types have their advantages. Monopropellant engines are more simple and can be scaled down to produce a thrust of less than 1N. They are favoured with reaction control systems (see chapter 4). On the other hand bipropellant pressure-fed systems have a higher specific impulse (around 300s compared with 230s) and so, in the interests of saving fuel, this system is usually used for the main engines aboard a spacecraft. For this reason it was the system initially considered as the primary propulsion for the mission

In the early stages of the design process, it was found that the mission required around 6300 m/s of ΔV from a low-Earth orbit to Titan insertion. Assuming that the launch vehicle would be able to take the spacecraft to an orbit between LEO and HCO, the estimated fuel mass fractions (equation 1.2) using the bipropellant system would be between 65% and 85%. Reducing this (rather high) range involves either finding a more fuel efficient sequence of maneuvers or finding a system with a greater specific impulse. With the assumption that the former wouldn't change significantly, the latter option was explored.

3.2 Ion Propulsion

Ion engines produce thrust by repelling an ion propellant at very high speeds with electrostatic (Coulomb) force. The propellant (typically xenon) is first ionised before it is accelerated through positive and negative grids resulting in a very high speed ion beam. To avoid building up a high negative charge on the spacecraft, a neutralising cathode fires electrons into the ion beam, restoring the propellant to a neutral state [1].

Thanks to the high velocity of the propellant ion-engines have and extremely high specific impulse - in general of the order of thousands of seconds. This allows for a much lower fuel requirement to achieve the same ΔV . Applying the same analysis as in the previous section to an ion thruster with a specific impulse of 3000s (typical value) gives fuel mass fractions between 10% and 20%. Besides the fact that this means a greatly lower fuel requirement, it also means that the mass of the fuel required is a lot less sensitive to the payload mass and the mission route. This is represented below:

Table 3.1: Sensitivity analysis of the two considered propulsion systems

I_{sp} [s]	ΔV [m/s]	m_d [kg]	m_t [kg]
3000	3000	1000	1107
	6300	1000	1239
	3000	1200	1329
	6300	1200	1486
310	3000	1000	2682
	6300	1000	7938
	3000	1200	3218
	6300	1200	9525

In the above table m_d is the dry mass value and m_t is the total mass. These estimates do not take into account the other propellants on-board. It is clear from this initial analysis that using the bipropellant system is both much more costly in terms of fuel but also much more sensitive to any changes in payload mass or ΔV requirement - leading to more unpredictable analysis. For these reasons, the ion propulsion system was chosen as the basis of the primary on-board propulsion.

While ion propulsion may be excellent in terms of efficiency it has major drawbacks in other areas - namely the low thrust output and the high electrical power demands. Due to the limitations in flow rate through the engine, the thrust is indeed incredibly low - generally of the order of $10^{-2}N$ compared with 10^2N for most spacecraft. This may seem too low to be practical, however, over a very long mission duration (roughly 8 years) a constant thrust, however small, will eventually build up a lot of speed. There are however limitations when working with such a low thrust which is where the secondary propulsion system will be used (see section 4.2). The orbital maneuvers are also more complicated. With more powerful engines the burn time is so small when compared with the orbital period is so small that the change in speed can be considered instantaneous. This is certainly not the case with ion engines where the burn time will be of a similar order of magnitude to the orbital period.

Lastly, the electrical power consumption of the engines is very high (generally between 0.5 and 10 kW). This is mostly problematic in the outer solar system where large solar panels are required to deal with the low solar radiation (see section 3.4).

3.3 NSTAR Thrusters

For this mission NASA Solar Technology Application Readiness (NSTAR) thruster was chosen for its excellent reliability. It has been used in two missions (Dawn and Deep Space 1) to date and has clocked up over 50,000 hours of burn time, more than any other rocket engine. The main specifications of the engine at maximum and minimum throttle are shown below:

Table 3.2: NSTAR thruster specifications [11]

	Maximum power	Minimum Power
Thrust [mN]	92	19
Power requirement [kW]	2.3	0.5
Specific impulse [s]	3000	1900

As discussed in the previous section there is a very high power requirement for only a very modest thrust. Due to the incredibly long burn times, it was decided that three of the thrusters would be used on-board the spacecraft - each operation one at a time. To ensure full, thrust vectoring, a gimbal was designed - this can be found in the appendix. The total dry weight of the system (not including tanks) is estimated to be 110kg.

3.4 Solar Power System

In space, electrical power either comes from radioisotope thermo-electric generators (RTGs) or solar panels. An early consideration (before considering ion propulsion) was to use one RTG (specifically the MMRTG) to power the base systems on-board the spacecraft as is typical for missions in the outer solar system (RTGs have the advantage of working regardless of distance from the sun). RTGs would be impractical for use with the ion propulsion, however, due to the relatively low power output (110W for the MMRTG). The remaining option, therefore, was solar power.

The power harvested by solar panels is described by the below equation:

$$P = \frac{L_0 A \eta_p}{R^2} \quad (3.1)$$

Where A is the surface area exposed to the sun (square meters), η_p is the efficiency, R is the distance from the sun (astronomical units) and L_0 is the solar intensity at 1AU. While L_0 and R are determined by physical/mission constraints, A and η_p are dependent on the design of the system.

The efficiency is determined by the solar technology. For the mission the Stretch Lens Array Square Rigger (SLASR) System was selected base on its lightweight, compact and efficient characteristics. It is currently in the final stages of testing and is predicted to be available well in advance of our mission launch. It has a BOL efficiency of 28% which, after natural degradation, reaches 22% after 9 years [12].

The surface area available depends on the design of the solar panels and how they are folded into a stowed configuration in the fairing before launch. The proposed design is to have two 21m 'wings' either side of the orbiter bus - each wing being composed of 7 3m by 3m panels, giving a total area of $126m^2$. To put that into context, the ISS has 16 solar arrays, each with $156m^2$ of area. Considering the thickness of the SLASR panels, each array can be folded to width of 490mm. The total weight of the arrays is predicted to be 130kg.

Using the values of area and efficiency, the available power over the voyage is shown below:

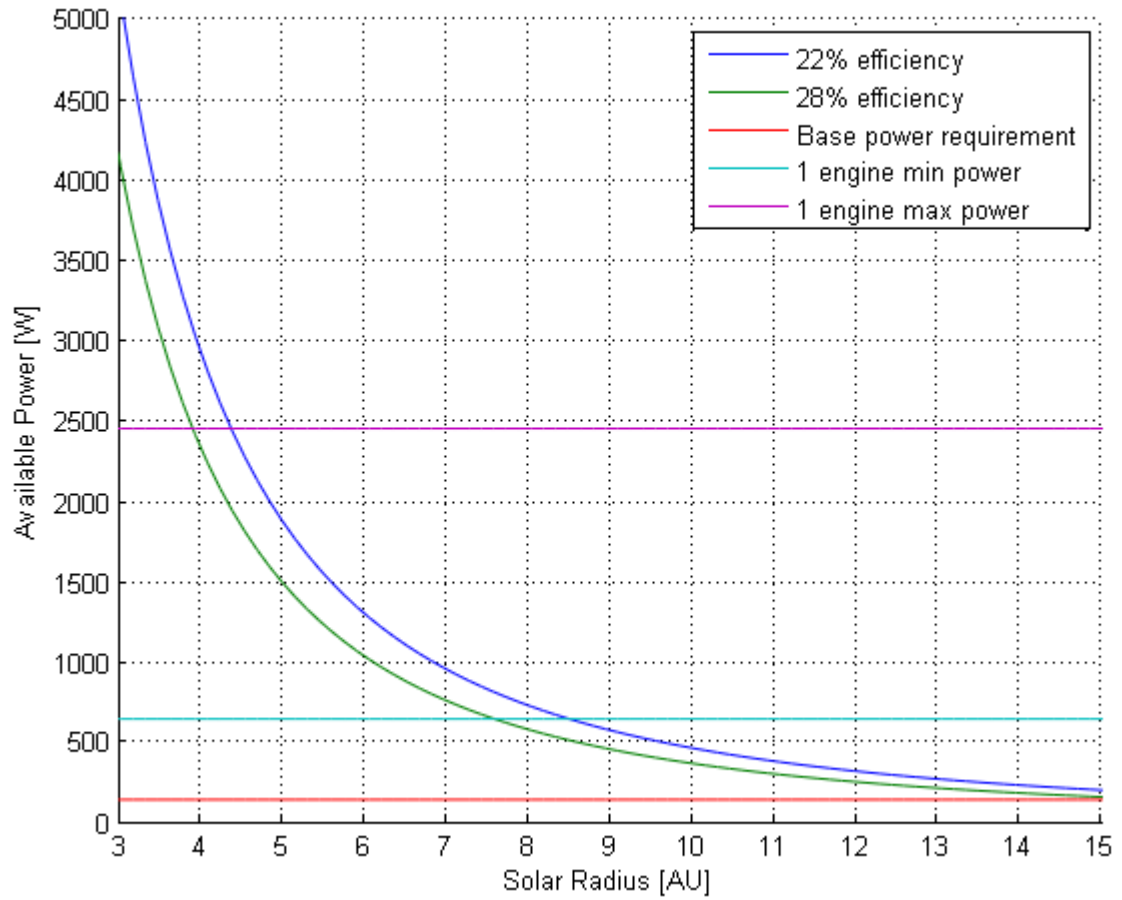


Figure 3.1: Available solar power vs distance from sun

The green curve is the BOL while the blue is the EOL. The vertical lines indicate fixed power requirement of the base systems as well as the engines. It is clear that between 4 and 4.5 AU the engines will no longer be operable at maximum power and will have to be shut down all together at around 8 AU. The Saturn system is located at 9.5 AU (on average). Therefore, while in-situ, the orbiter will have no issue with the base power demands as more than enough solar power is available. This information allowed the need for an expensive MMRTG (for use on the orbiter) to be eliminated.

Chapter 4

Reaction Control System and Secondary Propulsion

4.1 Attitude Control and RCS

There are three main ways of performing attitude adjustments in space: reaction wheels, thrust vectoring and reaction control systems. Thrust vectoring as already implemented in the main engine design, cannot provide complete attitude control nor can it work at all while the engines are shut down. A reaction control system produces rotation from the torque resultant of firing two thrusters either side of the center of mass. At least 12 thrusters are used for complete 3D rotation although up to 16 can be used for greater reliability.

Thrusters used in a RCS are universally monopropellant engines due to the very low thrust requirement. For our mission, a 1N-class hydrazine thruster manufactured by Airbus was chosen. Its main specifications are shown in table 4.1. The thrusters are located in clusters of 2 and 4 at the bottom of the spacecraft. a more detailed view of the thrusters can be found in [3].

4.2 Secondary Propulsion

The requirement for a secondary propulsion system was introduced in chapter 3. Additional engines are required for maneuvers which are beyond the scope of the main engines i.e. any maneuver too far from the sun or requiring a greater thrust than would otherwise be available. In order to limit the complexity of the fuel system, these thrusters also use hydrazine as the single propellant. The chosen engine was a 20N-class thruster, also manufactured by Airbus. Unlike the main engines, these do not require a gimbal for thrust vectoring as both engines will be operation simultaneously. The thrusters are located symmetrically either side of the main engines. The main specifications are listed below:

Table 4.1: Main specifications of monopropellant thrusters [13][14]

	RCS thrusters	Secondary thrusters
Propellant	hydrazine	hydrazine
Thrust [N]	1.1	24.6
Specific Impulse [s]	220	230
Dry mass [kg]	290	650
Length [mm]	172	195

Chapter 5

Propellant and Tank Mass

The total mass of the propellants and pressurant and their respective tanks must be calculated to provide a total mass value of the launch vehicle payload. In total there are three gases to be stored in spherical tanks: xenon, hydrazine and helium.

An initial estimate of the mass of xenon as a fraction of the total mass can be found using equation 1.2. This uses a ΔV value from the orbital maneuvers during the interplanetary voyage up until a distance of 8AU and a specific impulse value of 3000s - (the majority of the thrusting will be done within the maximum thrust operating range). The mass fraction of hydrazine was determined with the same equation. The ΔV value was estimated based on previous missions and a combination of the tasks that the monopropellant system was to perform (600m/s). Finally, the helium mass was estimated by finding a typical value of pressurant mass as a fraction of propellant mass - 0.3%

An initial estimate of the total mass of the spacecraft was found using the following equation. The mass fractions (m_{frac}) were simply the sum of the fuel mass fractions and the dry mass was taken to be the fixed masses (batteries, solar panels, Titan vehicle etc) plus an estimate of the tank masses.

$$m_{total} = \frac{m_{dry}}{1 - \sum m_{frac}} \quad (5.1)$$

This, in turn allows for the individual propellant masses to be calculated. Based on the densities of the stored propellants the internal volume can be calculated. The empty tank mass value comes from this volume as well as the material density and skin thickness. Finally, these values then provide a more accurate dry mass estimate which allows for an iterative process to converge on accurate values. In order to further the accuracy, *GMAT* was used to find the mass fractions. The results are summarised below:

Table 5.1: Tank and Propellant specifications

Tank	Propellant	Empty mass [kg]	Fuel mass [kg]	Internal volume [liters]
1	Xenon	9.3	161.9	82.6
2	Hydrazine	31.8	303.4	527.6
3	Helium	4.7	1.1	26.0

Chapter 6

Discussion

6.1 Total Mass

The final values for all the propulsion systems are listed below:

Table 6.1: Final mass values

Total mass [kg]	1603.5
Total propellant mass [kg]	531.4
Payload mass [kg]	420.0
Total empty tank mass [kg]	45.8
Primary propulsion dry mass [kg]	110.6
Secondary propulsion and RCS dry mass [kg]	4.9

6.2 Selection of Upper Stage Propulsion

As discussed in chapter 3 the motivation behind the selection of ion propulsion was the large mass savings that comes with the high specific impulse. Another concern was that the bipropellant system was very sensitive to changes in mass or orbital maneuvers - which could potentially lead to a redesign late in the design process. There was also the possibility that the mass would be too heavy or large for the Falcon 9 to carry - meaning that a larger, more expensive rocket would be selected in its place. In hindsight, however, these problems wouldn't have occurred and the bipropellant system would have given a fairly reasonable (although still significantly larger) total mass value. Had the design process continued for longer, this system could have been considered in more detail and its merit weighed against the ion system.

6.3 Selection of Ion Thruster

What was not mentioned in chapter 3 was the possibility of using other ion thrusters. This was mainly due to the vast operational time of the NSTAR system compared to others. However, there are some rival systems which are currently in development, notably the NASA Evolutionary Xenon Thruster (NEXT). Although still in development, the system shows potential with a better predicted performance than the NSTAR with a max I_{sp} of 4170s and a max thrust of 237mN. If these results were verified with flight time outside of the lab, the NEXT thruster would replace with NSTAR as the main engine in this mission [15].

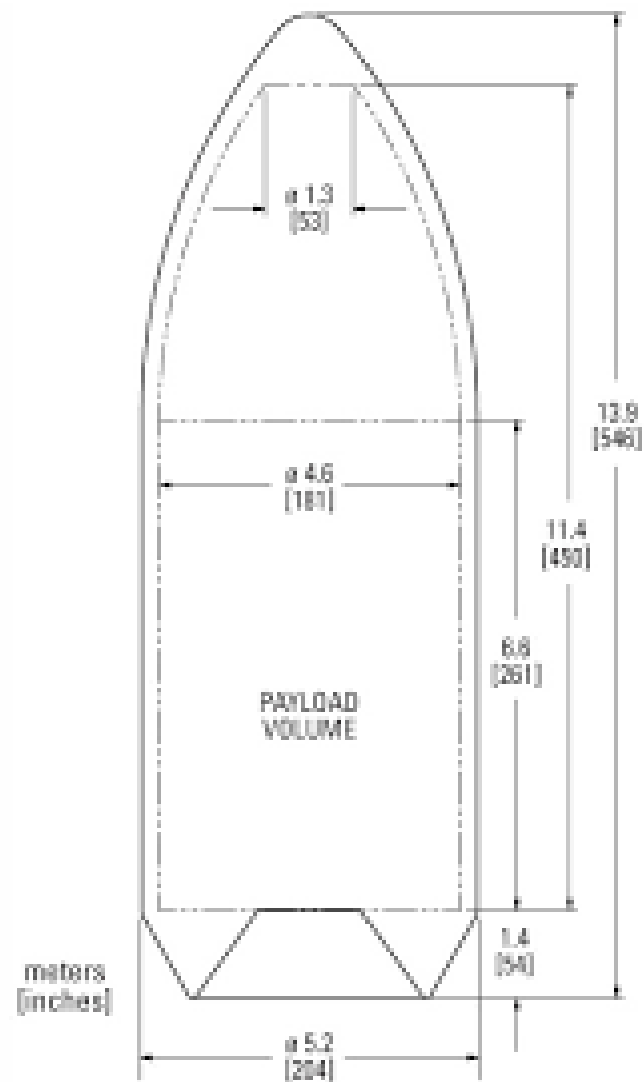
Bibliography

- [1] Sutton, G. Biblarz, O.: *Rocket Propulsion Elements*, 8th edition, 2010.
- [2] Poonja, R.: *Flyer and Orbiter Systems*, Imperial College London, 2010
- [3] Iqbal, U.: *Group Design Project Final Report: Titan Atmospheric Flyer FATT01*, Imperial college London, 2015
- [4] Dudley, J.: *Titan Atmospheric Flyer: Orbital Mechanics and Trajectory Design*, Imperial college London, 2015
- [5] *Delta IV Medium+*,
<http://www.spaceflight101.com/delta-iv-medium-52.html>,
Accessed 15/06/2015.
- [6] *Titan 4*,
<http://www.astronautix.com/lvs/titan4.htm>,
Accessed 15/06/2015.
- [7] *Atlas 5 Data Sheet*,
<http://www.spacelaunchreport.com/atlas5.html>,
Accessed 15/06/2015.
- [8] *Falcon 9*,
<http://www.spacex.com/falcon9>,
Accessed 15/06/2015.
- [9] *Falcon 9 v1.1 & F9R Launch Vehicle Overview*,
<http://www.spaceflight101.com/falcon-9-v11.html>,
Accessed 15/06/2015.
- [10] Chaikin, A.: *Is SpaceX Changing the Rocket Equation?*,
Air&Space Magazine, 2012
- [11] Brophy, JR. et al: *Ion Propulsion System (NSTAR) DS1 Technology Validation Report*,
Jet Propulsion Laboratory,
- [12] Benson, S.w.: *Solar Power for Outer Planets Study*,
NASA Glenn Research Center,
- [13] *1 N Chemical Monopropellant Hydrazine Thruster*,
<http://cs.astrium.eads.net/sp/spacecraft-propulsion/hydrazine-thrusters/1n-thruster.html>,
Accessed 15/06/2015.

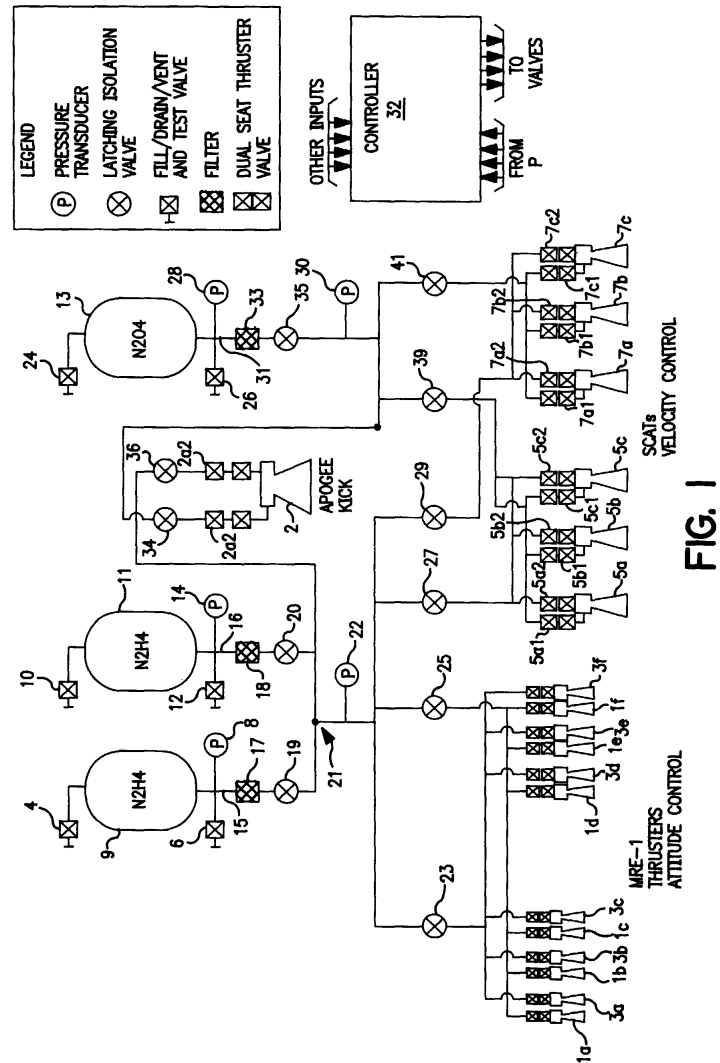
- [14] *20 N Chemical Monopropellant Hydrazine Thruster*,
<http://cs.astrium.eads.net/sp/spacecraft-propulsion/hydrazine-thrusters/20n-thruster.html>,
Accessed 15/06/2015.
- [15] Herman, D.A.: *NASA's Evolutionary Xenon Thruster (NEXT) Project Qualification Propellant Throughput Milestone: Performance, Erosion, and Thruster Service Life Prediction After 450 kg*,
NASA Glenn Research Center,
- [16] bassichis, J, Hook, D, Sackheim, R: *Spacecraft attitude and velocity control thruster system*,
1999,

Appendix A

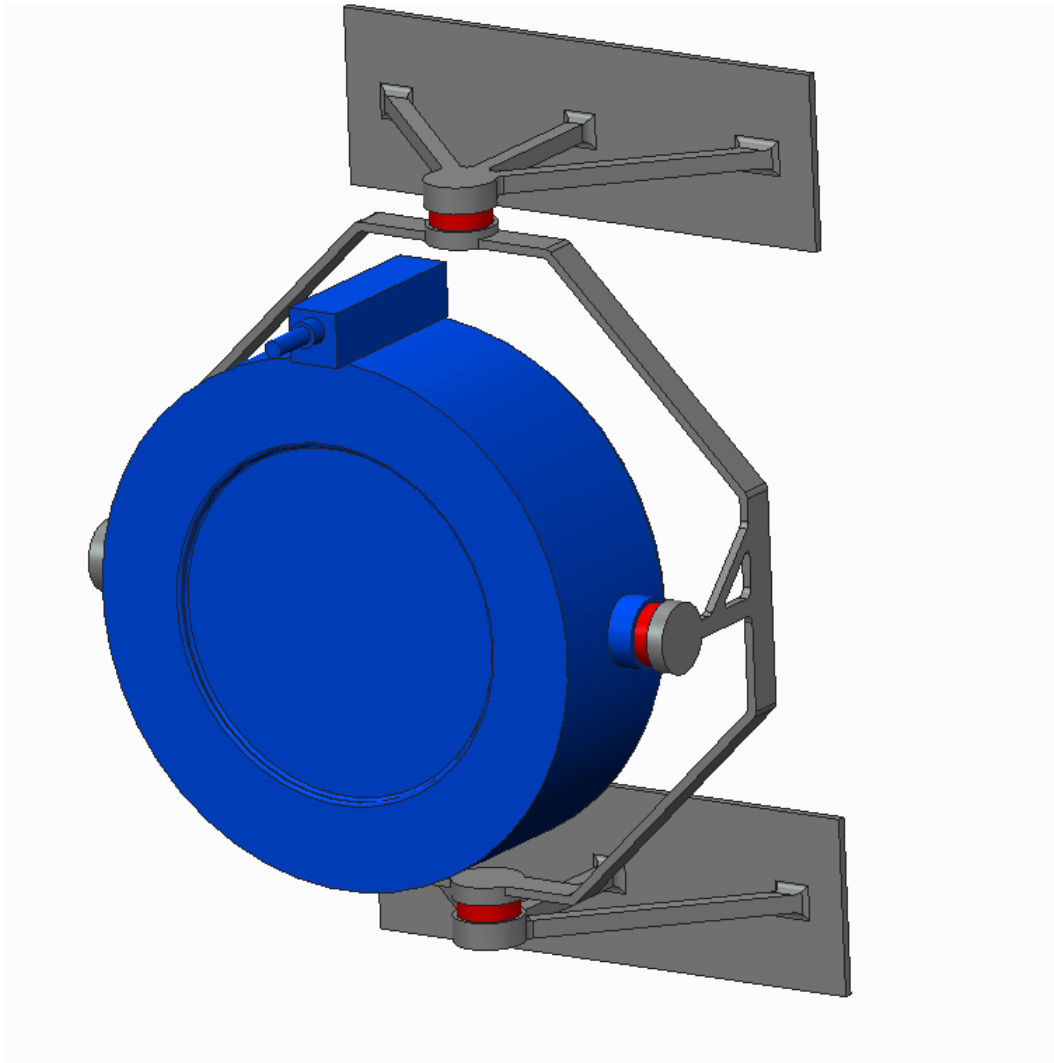
A.1 Falcon 9 Payload Fairing



A.2 Bipropellant Propulsion System



A.3 Main Engine Gimbal Design



A.4 Launch Vehicle Comparison

